

Data Mining Chapter 3 Linear Regression Group 1

Ishimwe Aime Cesar 101416

August 2026

Question 1

Describe the null hypotheses to which the p-values given in Table 3.4 correspond. Explain what conclusions you can draw based on these p-values. Your explanation should be phrased in terms of sales, TV, radio, and newspaper, rather than in terms of the coefficients of the linear model.

Answer

- Null hypothesis

$$(H_0) : B_{TV} = B_{radio} = B_{newspaper} = 0$$

- Conclusions are drawn from p-values:

- There is a relation between advertising sales and budget.
- The p values for TV and radio are statistically significant but for newspaper it is not significant in presence of the TV and radio.
- So we can make a Multiple linear regression model with only TV and Radio predictors and drop newspaper.

Question 6

Using (3.4), argue that in the case of simple linear regression, the least squares line always passes through the point $(\bar{x}, \bar{y})$.

Answer

$$\hat{y}_i = \hat{B}_0 + \hat{B}_1 * x$$

when: $x = \bar{x}$

$$\hat{B}_0 = \bar{y} - \hat{B}_1 * \bar{x}$$

$$\hat{y}_i = \bar{y} - \hat{B}_1 * \bar{x} + \hat{B}_1 * \bar{x} = \bar{y}$$

Question 11

In this problem, we will investigate the t -statistic for the null hypothesis

$$H_0 : \beta = 0$$

in simple linear regression without an intercept. To begin, we generate a predictor x and a response y as follows.

```
rng = np.random.default_rng(1)
x = rng.normal(size=100)
y = 2 * x + rng.normal(size=100)
```

- (a) Perform a simple linear regression of y onto x , *without* an intercept. Report the coefficient estimate $\hat{\beta}$, the standard error of this coefficient estimate, and the t -statistic and p -value associated with the null hypothesis

$$H_0 : \beta = 0.$$

Comment on these results. (You can perform regression without an intercept using the keyword argument `intercept=False` to `ModelSpec()`.)

Answer

We perform a simple linear regression of Y onto X without an intercept.

The fitted model gives:

$$\hat{\beta} = 2.1067$$

with standard error

$$SE(\hat{\beta}) = 0.106.$$

The corresponding t -statistic is

$$t = 19.792,$$

and the p -value is approximately

$$p \approx 0.$$

Since the p -value is extremely small, we reject the null hypothesis

$$H_0 : \beta = 0.$$

Therefore, there is strong evidence that X has a significant linear relationship with Y .

- (b) Now perform a simple linear regression of x onto y without an intercept, and report the coefficient estimate, its standard error, and the corresponding t -statistic and p -values associated with the null hypothesis

$$H_0 : \beta = 0.$$

Comment on these results.

Answer

We now perform a simple linear regression of X onto Y without an intercept.

The fitted model gives:

$$\hat{\beta} = 0.3789$$

with standard error

$$SE(\hat{\beta}) = 0.019.$$

The corresponding t -statistic is

$$t = 19.792,$$

and the p -value is approximately

$$p \approx 0.$$

Again, because the p -value is extremely small, we reject

$$H_0 : \beta = 0.$$

Thus, there is strong evidence of a significant linear relationship between X and Y .

- (c) What is the relationship between the results obtained in (a) and (b)?

Answer

The regressions in parts (a) and (b) have different coefficient estimates and standard errors. However, they produce the same t -statistic and hence the same p -value.

For the regression of Y onto X ,

$$\hat{\beta} = 2.1067,$$

whereas for the regression of X onto Y ,

$$\hat{\beta} = 0.3789.$$

Nevertheless, both regressions have

$$t = 19.792.$$

Consequently, both regressions lead to the same conclusion concerning the null hypothesis $H_0 : \beta = 0$.

The original relationship can be written approximately as

$$Y = 2X + \epsilon,$$

which can also be rearranged conceptually as

$$X = \frac{1}{2}(Y - \epsilon).$$

- (d) For the regression of Y onto X without an intercept, the t -statistic for

$$H_0 : \beta = 0$$

takes the form

$$\frac{\hat{\beta}}{SE(\hat{\beta})},$$

where $\hat{\beta}$ is given by (3.38), and where

$$SE(\hat{\beta}) = \sqrt{\frac{\sum_{i=1}^n (y_i - x_i \hat{\beta})^2}{(n-1) \sum_{i'=1}^n x_{i'}^2}}.$$

(These formulas are slightly different from those given in Sections 3.1.1 and 3.1.2, since here we are performing regression without an intercept.) Show algebraically, and confirm numerically in R, that the t -statistic can be written as

$$\frac{(\sqrt{n-1}) \sum_{i=1}^n x_i y_i}{\sqrt{\left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i'=1}^n y_{i'}^2 \right) - \left(\sum_{i'=1}^n x_{i'} y_{i'} \right)^2}}.$$

Answer

For a regression of Y onto X without an intercept, the t -statistic is

$$t = \frac{\hat{\beta}}{SE(\hat{\beta})}.$$

For regression without an intercept,

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}.$$

Also,

$$SE(\hat{\beta}) = \sqrt{\frac{\sum_{i=1}^n (y_i - x_i \hat{\beta})^2}{(n-1) \sum_{i=1}^n x_i^2}}.$$

Substituting these expressions into the formula for the t -statistic gives

$$t = \frac{\frac{\sum_{i=1}^n x_i y_i}{\sum_{j=1}^n x_j^2}}{\sqrt{\frac{\sum_{i'=1}^n (y_{i'} - x_{i'} \hat{\beta})^2}{(n-1) \sum_{j'=1}^n x_{j'}^2}}}.$$

Simplifying,

$$t = \frac{\sqrt{n-1} \sum_{i=1}^n x_i y_i}{\sqrt{\left(\sum_{j=1}^n x_j^2 \right) \left(\sum_{i'=1}^n (y_{i'} - x_{i'} \hat{\beta})^2 \right)}}.$$

Now consider the quantity in the denominator:

$$\left(\sum_{j=1}^n x_j^2 \right) \left(\sum_{i'=1}^n (y_{i'} - x_{i'} \hat{\beta})^2 \right).$$

Expanding the squared term,

$$= \left(\sum_{j=1}^n x_j^2 \right) \left[\sum_{i'=1}^n \left(y_{i'}^2 - 2y_{i'} x_{i'} \hat{\beta} + x_{i'}^2 \hat{\beta}^2 \right) \right].$$

Using

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2},$$

the expression simplifies to

$$\left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right) - \left(\sum_{i=1}^n x_i y_i \right)^2.$$

Therefore,

$$t = \frac{\sqrt{n-1} \sum_{i=1}^n x_i y_i}{\sqrt{\left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right) - \left(\sum_{i=1}^n x_i y_i \right)^2}}$$

which is the required result.

Using the data numerically gives

$$t = 19.7918,$$

which agrees with the regression output in part (a).

- (e) Using the results from (d), argue that the t-statistic for the regression of y onto x is the same as the t-statistic for the regression of x onto y.

Answer

From part (d), the t -statistic can be expressed as

$$t = \frac{\sqrt{n-1} \sum_{i=1}^n x_i y_i}{\sqrt{\left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right) - \left(\sum_{i=1}^n x_i y_i \right)^2}}.$$

If we reverse the regression and regress X onto Y , the expression becomes

$$t = \frac{\sqrt{n-1} \sum_{i=1}^n y_i x_i}{\sqrt{\left(\sum_{i=1}^n y_i^2 \right) \left(\sum_{i=1}^n x_i^2 \right) - \left(\sum_{i=1}^n y_i x_i \right)^2}}.$$

Since

$$x_i y_i = y_i x_i$$

and multiplication is commutative,

$$\left(\sum x_i^2 \right) \left(\sum y_i^2 \right) = \left(\sum y_i^2 \right) \left(\sum x_i^2 \right),$$

the two expressions are identical.

Therefore, the t -statistic for the regression of Y onto X is the same as the t -statistic for the regression of X onto Y .

Numerically, both give approximately

$$t = 19.792.$$

- (f) In R, show that when regression is performed *with* an intercept, the t -statistic for

$$H_0 : \beta_1 = 0$$

is the same for the regression of y onto x as it is for the regression of x onto y .

Answer

Finally, we perform the two regressions with an intercept.

For the regression of Y onto X , the fitted model gives

$$\hat{\beta}_0 = 0.1470$$

and

$$\hat{\beta}_1 = 2.0954.$$

The standard error of the slope coefficient is

$$SE(\hat{\beta}_1) = 0.106,$$

and the t -statistic for testing

$$H_0 : \beta_1 = 0$$

is

$$t = 19.783.$$

For the regression of X onto Y , the fitted model gives

$$\hat{\beta}_0 = -0.0440$$

and

$$\hat{\beta}_1 = 0.3817.$$

The standard error of the slope coefficient is

$$SE(\hat{\beta}_1) = 0.019,$$

and the corresponding t -statistic is again

$$t = 19.783.$$

Therefore,

$$t_{Y \text{ on } X} = t_{X \text{ on } Y} = 19.783$$

even when an intercept is included.

Hence, the numerical results confirm that the t -statistic for testing

$$H_0 : \beta_1 = 0$$

is the same whether we regress Y onto X or X onto Y .